

SQL PRACTICE TASKS

Topics Covered : VIEWS • JOINS • INDEXES

Database: MySQL (XAMPP)

♦ COMMON TABLE SETUP (Use for all tasks)

Copy code

Sql

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    city VARCHAR(50),  
    age INT  
);  
  
CREATE TABLE courses (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(50),  
    fees INT  
);  
  
CREATE TABLE enrollments (  
    enroll_id INT PRIMARY KEY,  
    student_id INT,  
    course_id INT,  
    enroll_date DATE,  
    FOREIGN KEY (student_id) REFERENCES students(student_id),  
    FOREIGN KEY (course_id) REFERENCES courses(course_id)  
);
```

♦ Sample Data

Copy code

Sql

```
INSERT INTO students VALUES  
(1,'Rahul','Ahmedabad',21),  
(2,'Priya','Surat',22),
```

```
(3,'Amit','Vadodara',20),  
(4,'Neha','Ahmedabad',23);
```

INSERT INTO courses VALUES

```
(101,'Java',15000),  
(102,'Python',12000),  
(103,'Web Development',10000);
```

INSERT INTO enrollments VALUES

```
(1,1,101,'2024-01-10'),  
(2,2,102,'2024-01-12'),  
(3,3,101,'2024-01-15'),  
(4,1,103,'2024-01-20');
```

TASKS

TASK 1: INNER JOIN

Task:

Display student name, course name, and enrollment date.

 Concept: INNER JOIN

TASK 2: LEFT JOIN

Task:

Display all students with their enrolled course names.

Show NULL if a student is not enrolled.

 Concept: LEFT JOIN

TASK 3: RIGHT JOIN

Task:

Display all courses and students enrolled in them.

 Concept: RIGHT JOIN

TASK 4: JOIN + WHERE

Task:

Display students who are enrolled in the Java course.

✚ Concept: JOIN + WHERE

✓ TASK 5: CREATE VIEW

Task:

Create a view named `student_course_view`
that shows student name, city, course name, and fees.

✚ Concept: VIEW

✓ TASK 6: SELECT FROM VIEW

Task:

Fetch all records from `student_course_view`
where city = 'Ahmedabad'.

✚ Concept: VIEW + WHERE

✓ TASK 7: UPDATE THROUGH VIEW

Task:

Update city to 'Gandhinagar' for student Rahul using the view.

✚ Concept: Updatable VIEW

✓ TASK 8: CREATE INDEX

Task:

Create an index on city column of students table.

✚ Concept: INDEX

✓ TASK 9: COMPOSITE INDEX

Task:

Create a composite index on `student_id` and `course_id` in enrollments table.

✚ Concept: Composite INDEX

✓ TASK 10: PERFORMANCE CHECK

Task:

Use EXPLAIN to analyze the query that fetches students from Ahmedabad.

✚ Concept: INDEX + EXPLAIN

Copy code

Sql

```
SELECT * FROM students WHERE city='Ahmedabad';
```